

Jian-Qing Qi

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RESEARCH INTERESTS

Physical Organic Chemistry, Photochemistry, Radical Chemistry.

EDUCATION

Tsinghua University, Department of Chemistry

Aug. 2022–Now

PhD Student (Adviser: Lei Jiao)

Tsinghua University, Department of Chemistry

Sept. 2018–July. 2022

Bachelor of Science (Graduate with Honors)

RECENT RESEARCH

Direct observation of all open-shell intermediates in a photocatalytic cycle

March 2024

J. Am. Chem. Soc. **2024**, 146 (11), 7140–7145.

- Successfully achieved direct observation of all radical intermediates in a photocatalytic cycle, clearly characterizing their interconversion processes.
- Marking the first-time chemists can clearly observe all radical intermediates and their transformation processes in a complex reaction
- Providing a new perspective for understanding photocatalytic reaction mechanisms.

Overestimated halogen atom transfer reactivity of α -aminoalkyl radicals

August 2024

J. Am. Chem. Soc. **2024**, 146 (37), 25860–25869.

- In-depth study of the reactivity of amine alkyl radicals in halogen atom transfer reactions, and established a systematic reactivity scale
- Challenges the conventional understanding of amine alkyl radical reactivity and demonstrates the tremendous potential of U-PSD TREPR in unveiling radical kinetics.

Pyridinium(boron) fluorophores as versatile photocatalysts

June 2026

J. Am. Chem. Soc. **2026**. Accepted

- New photocatalyst fluorophore featuring facile synthesis, high modularity and turnability.
- Photo-redox catalysis and energy-transfer catalysis.
- In-depth analysis of the photophysical and photochemical properties of this fluorophore, and its structure-reactivity relationship.

HONORS & AWARDS

- June 2025. **Young Scientist Nominee** of Lindau Nobel Laureate Meetings.
 - August 2024. **Selected Oral** at the Excellent Doctoral Student Forum of the National High-Level Talent Training Center for Chemistry.
 - April 2024. **Outstanding Presenter Award** at 5th Beijing-Tianjin Organic Chemistry Graduate Student Academic Seminar.
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SELECTED PUBLICATION

- (1) Bai, L. [#]; **Qi, J.-Q.** [#]; Liu, X.; Yang, J.^{*}; Guo, X.^{*}; Jiao, L.^{*} Pyridinium(boron) Fluorophores as Versatile Photocatalysts. *J. Am. Chem. Soc.* **2026**, Accepted.
- (2) Suo, W. [#]; **Qi, J.-Q.** [#]; Liu, J.; Sun, S.; Jiao, L.^{*}; Guo, X.^{*} Overestimated Halogen Atom Transfer Reactivity of α -Aminoalkyl Radicals. *J. Am. Chem. Soc.* **2024**, *146* (37), 25860–25869.
- (3) **Qi, J.-Q.** [#]; Suo, W. [#]; Liu, J.; Sun, S.; Jiao, L.^{*}; Guo, X.^{*} Direct Observation of All Open-Shell Intermediates in a Photocatalytic Cycle. *J. Am. Chem. Soc.* **2024**, *146* (11), 7140–7145.
- (4) Zhang, S. [#]; Cheng, L. [#]; **Qi, J.-Q.** [#]; Jia, Z.; Zhang, L.; Jiao, L.^{*}; Guo, X.^{*}; Luo, S.^{*} Characterization and Monitoring of Transient Enamine Radical Intermediates in Photoredox/Chiral Primary Amine Synergistic Catalytic Cycle. *CCS Chemistry* **2024**, *6* (10), 2420–2426.
- (5) Zhang, S.; Zhou, S.; **Qi, J.**; Jiao, L.; Guo, X. Time-Resolved Electron Paramagnetic Resonance Spectrometer Based on Ultrawide Single-Sideband Phase-Sensitive Detection. *Rev. Sci. Instrum.* **2023**, *94* (8), 084101.
- (6) **Qi, J.-Q.**; Jiao, L. DFT Study on the Mechanism of 4,4'-Bipyridine-Catalyzed Nitrobenzene Reduction by Diboron(4) Compounds. *J. Org. Chem.* **2020**, *85* (21), 13877–13885.